
Design of RF and mm-Wave Integrated Circuits

An Open-Source Summary

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1 Motivation

We want to go to higher frequency to transmit more and more data as per Shannon channel capacity

$$BW \cdot \log_2 \left(1 + \frac{S}{N} \right)$$

1.1 Problem 1

It was predicted that the f_{max} and f_T of CMOS would increase in the coming years. Sadly, this is not true as CMOS won't go much higher than 300 GHz. One of our best hope is **Si-Ge** which is a developing and promising technology for high speed application. The main drawback is the fact it doesn't scale well (factor 1000 between CMOS and Si-Ge) and it is a power hungry technology node.

1.2 Problem 2

Another big problem comes from fundamental EM theory. The **free-space path loss**:

$$20 \log_{10} \frac{4\pi f_c d}{c}$$

The higher we go in frequency, the higher is the losses. Same with the Friis equation:

$$P_{\text{density}} = \frac{P_t}{4\pi d^2} \cdot G_t \quad (1)$$

$$P_r = P_{\text{density}} \cdot \frac{\lambda^2}{4\pi} \cdot G_r \quad (2)$$

$$\frac{P_r}{P_t} = \left(\frac{\lambda}{4\pi d} \right)^2 \cdot G_r \cdot G_t \quad (3)$$

But, the gain of an antenna scales with the frequency. Which means, for the same area of a RF one, we can split it in multiple sub mmWave one.

$$G_{\text{ant}} = \frac{4\pi A_{\text{ant}} f_c^2}{c^2}$$

This opens the way to *beamforming* antenna where tiny delays can steer the beam. Main issue is the energy consumption, it has currently troubles to find its way to the consumer market. Many architecture exists that try to overcome various challenges.

2 mm-wave design in CMOS: Actives

2.1 MOS transistor power gain, fT and fMAX

2.2 Gate resistance

2.3 Stability

2.4 Neutralization

2.5 Drawback of Neutralization

2.5.1 CM stability

2.5.2 CM feedback

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3.1 Modification

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